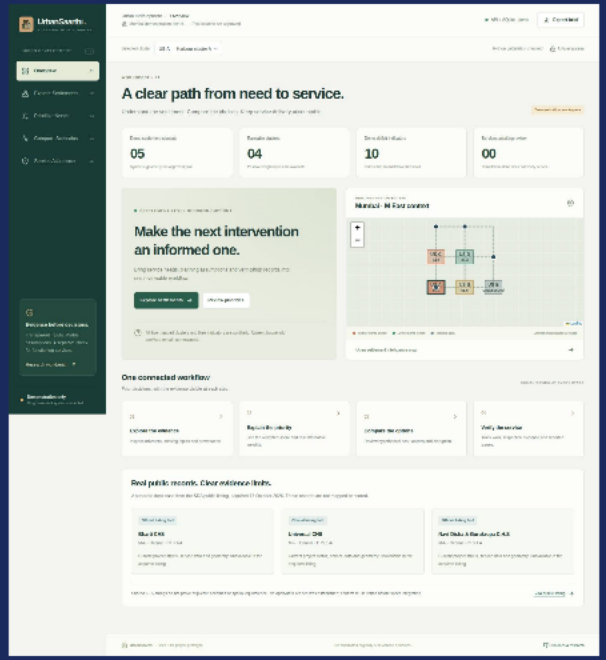
UrbanSaarthi

Explainable settlement planning
and basic-service assurance



Working software prototype
Mumbai demonstration context
No approved field pilot or measured resident impact

2

The Problem

Proposed task: the next water-service inspection in one agreed M-East cluster

What is the problem?

A work can be physically complete while reliable service still needs checking. Reviewers need dated evidence for the next inspection and closure decision.

Who is affected, and where?

Prospective users: BMC M-East water-service reviewers, asset operators and residents. No operational cluster or agency partner is confirmed.

Evidence

Mumbai water research documents reliability concerns. Three acquired SRA listing rows supply locality leads, with no acquired service observations.

Current gaps

SRA GIS, SBM monitoring and Swachhata already exist. Local omissions, review effort and the benefit of this workflow still need testing.

Unmeasured

Current affected-household count in the proposed pilot area

Why it matters

An unverified closure may leave a service problem unresolved.

3 Our Solution & Innovation

One reviewable workflow, with assumptions and unknowns visible

Our solution in brief

Five screens connect a settlement map, explainable priority scores, hypothetical intervention options and a separate service-evidence review.

How it works

1 Input

Dated indicators and source records

2 Process

Transparent scores and scenario costs

3 Output

Inspection brief and service review

What makes it new?

	Existing approach	UrbanSaarthi
Cost	*Not measured*	*Assumed INR 132,000*
Speed / time	*Not measured*	*Paired test planned*
Reach / access	*Existing channels*	*Five demo clusters*
Ease of use	*Not measured*	*User testing needed*

Proposed contribution

Traceability across planning and verification. Incremental value remains to be tested against the current process.

4 Feasibility & Implementation

Current stage: working prototype. Field validation and operating ownership remain pending.

Current stage: Working Prototype



Technology

React / Leaflet
FastAPI / SQLite
Deterministic scoring
Python consistency checks

Synthetic service inputs

Estimated cost

INR 132,000 assumed
20 asset review scopes
INR 6,600 / scope

Household cost unknown
Construction excluded

Timeline

3 months: bounded trial
6 months: second-site test
12 months: conditional handover

Each step needs evidence and approval

Partners

Prospective: BMC M-East ward and Assistant Engineer, Water Works

No office, operator or partner has agreed

5

Impact & Scalability

Intended outcomes require an independent comparison and repeat observations

Unmeasured

Beneficiaries reached

No verified beneficiary denominator

Unmeasured

Saving in cost / time

Paired review comparison proposed

Unmeasured

Jobs or livelihoods

Employment-hub distance is a demo proxy

Social impact

Test critical evidence omissions, review effort and repeat service observations. Include voluntary assisted feedback and disability/access barriers.

Sustainability

Illustrative continuation: INR 162,360 / year. A recurring payer and maintenance owner remain unconfirmed.

Scale-up plan

Validate one task, then a second ward. Expansion needs permitted data, authenticated roles, approved integration and operating finance.

6 Team/Individual & Support Required

Individual participant. Personal and eligibility details will be completed privately.

Individual applicant

Working research-based prototype

Personal details omitted at applicant request
No operational users or agency endorsement

Mentor and independent field partner: unconfirmed

Contact and eligibility details: complete privately

Support we are seeking

- [ ] Mentorship & technical guidance
- [ ] Pilot with a government department
- [ ] Funding / grant support
- [ ] Incubation & workspace
- [ ] Testing, certification or IP support

A bounded review task, permitted records and independent field assessment before expansion.